

Program Nexp_GLSE

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1. Introduction.

Nexp_GLSE is a Global Least Square Estimator for fitting lifetime or dwell time distributions by a linear combination of N exponential terms. The experimental setup is assumed to have an infinitely narrow Impulse Response Function (IRF), therefore no convolution is carried out. Other programs by the same author are designed specifically for the experimental data with wide IRF, for example, for Time Correlated Single Photon Counting (TCSPC) data; there convolution is included in the model function. The weighting of squared residuals is based on the assumption of Poissonian distribution of the counts in the bins. The program can handle bins of equal or unequal widths. The program can analyze multiple data sets simultaneously, with some of the model parameters being shared between two or more data sets (global parameters), while the other model parameters being relevant to only one data set (local parameters). The bin widths can be different for different data sets. Minimization of weighted least squares is performed using a version of Levenberg-Marquardt algorithm.

2. Model function.

Two different model functions can be considered: the survival function $S(t)$ and the probability density function $f(t)$; the argument t is time in both cases. The survival function $S(t)$ predicts how many of the experimentally detected lifetimes / dwell times are longer than t . The probability density function $f(t)$ predicts the number of lifetimes or dwell times between t and $t+dt$, divided by dt , where dt is a very small time interval. From these definitions it follows that

$$f(t) = - \frac{d}{dt} S(t) \quad (1)$$

The most common approximations for $S(t)$ and $f(t)$ are linear combinations of exponential terms,

$$S(t) = \sum_{i=1}^{N_{\text{exp}}} A_i \exp(-t/\tau_i) = \sum_{i=1}^{N_{\text{exp}}} A_i \exp(-k_i t) \quad (2)$$

$$f(t) = \sum_{i=1}^{N_{\text{exp}}} \alpha_i \exp(-t/\tau_i) = \sum_{i=1}^{N_{\text{exp}}} \alpha_i \exp(-k_i t) \quad (3)$$

where N_{exp} is the number of exponential terms, A_i and α_i are the amplitudes or pre-exponential factors, τ_i are the decay times, and k_i are the decay rates. The relation between k_i and τ_i is obvious:

$$k_i = 1/\tau_i \quad (4)$$

Using equation (1) one can also derive the following relation between the uppercase and lowercase alphas:

$$\alpha_i = A_i/\tau_i \quad (5)$$

Normalization of alphas is the next question. If the survival function $S(t)$ is defined as the probability that a lifetime / dwell time is greater than t , then $S(0)$ must be equal to 1, from which it follows that the sum of all A_i must be equal to 1, which results in a normalization condition. If the sum of all A_i is not equal to 1, then the amplitudes can be normalized by dividing each A_i by the sum of all A_i ; the amplitudes obtained in this way will be referred to as the *fractional* A_i :

$$\text{fractional } A_i = \frac{A_i}{\sum_{j=1}^{N_{\text{exp}}} A_j} \quad (6)$$

However, if the survival function $S(t)$ is defined as the number of lifetimes / dwell times greater than t , then $S(0)$ must be close to the total number of lifetimes / dwell times recorded during the experiment. This definition is more convenient when the model function from equation (2) is used to fit experimental data. The amplitudes A_i that are not normalized to a unit sum play the roles of the actual fitting parameters; to distinguish them from the *fractional* A_i we will also call them the *absolute* A_i . Likewise, the lowercase alphas that are calculated using equation (5) will be referred to as the *absolute* α_i ; these *absolute* α_i can be also normalized to unit sum, which will result in *fractional* α_i :

$$\text{fractional } \alpha_i = \frac{A_i / \tau_i}{\sum_{j=1}^{N_{\text{exp}}} A_j / \tau_j} \quad (7)$$

3. Binning of lifetimes / dwell times.

The bins are defined in the following manner. Bin number 1 begins at $t=0$ and ends at $t=t_1$. Bin number 2 begins at $t=t_1$ and ends at $t=t_2$. Bin number n begins at $t=t_{n-1}$ and ends at $t=t_n$. The bin boundaries t_n can be equally-spaced or not equally spaced, but they must be all positive and they must represent a monotonically increasing sequence, i. e., $t_n < t_{n+1}$. If the recorded lifetimes / dwell times are digitized with a finite discretization step, then all t_n must be multiples of that discretization step; the significance of this condition will be illustrated using an example below. Other than that, the choice of the times t_n is completely free. All experimental lifetimes / dwell times t must be positive numbers (cannot be zero or negative). A lifetime / dwell time t is placed in such bin n that $t_{n-1} < t \leq t_n$. The count of lifetimes / dwell times in bin n will be denoted D_n . The integers D_n follow Poissonian statistics. The set of all D_n , where $n=1,2,\dots$ represents one experimental data set.

The theoretical model functions introduced in equations (2) and (3) do not represent the numbers of counts in the bins, however, the theoretical prediction for the numbers of counts in the bins can be expressed in terms of the function $S(t)$ or $f(t)$:

$$M_n = S(t_{n-1}) - S(t_n) \quad (8)$$

$$M_n = \int_{t_{n-1}}^{t_n} f(t) dt \quad (9)$$

It is easier to use equation (8) than equation (9), therefore the program Nexp_GLSE uses the function $S(t)$ rather than the function $f(t)$ to calculate the theoretical prediction for the numbers of counts in the bins; the actual model parameters are the capital alphas A_i and the decay times τ_i .

Now consider the following example emphasizing the effect of the discretization step Δt . If the experimental setup has a discretization step $\Delta t = 10$ ms and we choose equally-spaced bins of 15 ms width, then the first bin will contain only 10 ms dwell times, the second bin will contain 20 ms and 30 ms dwell times, the third bin will contain only 40 ms dwell times, the fourth bin will contain 50 ms and 60 ms dwell times, etc.. In other words, every even-numbered bin will contain about twice as many dwell times as a nearby odd-numbered bin. This should not be the case if the odd and even bins have the same width. However, when the bin boundaries t_n are not exact multiples of the discretization step Δt , the de facto bin widths are not the same as the theoretically defined bin widths. The program Nexp_GLSE assumes that the theoretically defined bin widths are always correct. Therefore, to avoid serious problems, one should always make the bin boundaries t_n exact multiples of the discretization step Δt . If, however, the discretization step Δt is extremely small as compared to the bin width, then this requirement can be disregarded.

4. The importance of correct weighting and the definition of the χ^2 .

In general, a (weighted) least squares technique searches for a set of parameters that minimizes the sum

$$SSQ = \sum_{n=1}^{N_D} W_n [D_n - M_n(\vec{p})]^2 \quad (10)$$

where N_D is the total number of data points, i. e., the number of bins in our case; W_n is the weight for bin n , D_n is the count of dwell times in bin n , $M_n(\vec{p})$ is the model prediction for bin n , which depends on the parameter vector $\vec{p}$; the latter is a set of all A_i and τ_i as defined above.

The choice of the weights W_n is important because it affects the accuracy of the parameter estimates $\vec{p}$ obtained by minimizing the SSQ. Walter Clark Hamilton defined the most accurate parameter estimates as those estimates that have the least possible root-mean-square deviations from the true values of these parameters, and he also proved¹ that the most accurate parameter estimates are obtained when the weights W_n are equal to the inverse variances in the data D_n :

$$W_n = \frac{1}{\text{Var}(D_n)} \quad (11)$$

There is one additional advantage of using the weights defined in equation (11): with this weighting the SSQ defined in equation (10) becomes identical to the χ^2 defined as

$$\chi^2 = \sum_{n=1}^{N_D} \frac{[D_n - M_n(\vec{p})]^2}{\text{Var}(D_n)} \quad (12)$$

This makes it possible to apply the chi-squared goodness-of-fit test directly to the result of the minimization of the sum of the weighted least squares. This test helps to answer the question whether the model adequately fits the data or not. This is done by comparing the minimized χ^2 value with tables of $\chi^2(P, N_F)$, where P is the probability that the actual χ^2 value does not exceed $\chi^2(P, N_F)$, and N_F is the number of degrees of freedom, calculated as the difference $N_D - N_P$, where N_P is the number of model parameters whose values are estimated from the same data set (N_P does not include the parameters that were known *a priori*). A fit of the model to the data is usually considered adequate if the minimized χ^2 from equation (12) does not exceed $\chi^2(0.95, N_F)$. If the minimized χ^2 exceeds $\chi^2(0.995, N_F)$, then the model clearly must be rejected. For the ease of comprehension, instead of the absolute χ^2 defined in equation (12) it is common to display the value of the *reduced* χ^2 , defined as the ratio

$$\text{reduced } \chi^2 = \frac{\chi^2}{N_F} \quad (13)$$

For a good fit the *reduced* χ^2 value is expected to be close to 1. If the *reduced* χ^2 value is less than 1, then it is not only a good fit, but it is also a lucky data set. The program Nexp_GLSE outputs not only the minimized *reduced* χ^2 value for the given data set and the given model, but it also

1 Walter Clark Hamilton *Statistics in physical science: Estimation hypothesis testing, and least squares* Ronald Press, New York, 1964

outputs the values of *reduced* $\chi^2(0.95, N_F)$, *reduced* $\chi^2(0.995, N_F)$, the probability P such that the $\chi^2(1-P, N_F)$ equals the minimized χ^2 value from equation (12), the minimized χ^2 value, and the Akaike Information Criterion (AIC), which is used to choose between models with different N_p .

5. Accurate estimation of variances for binned lifetime / dwell time data.

Since the counts D_n represent the numbers of events that are completely independent of one another (each dwell time is independent of any other dwell time registered before or after it), the numbers D_n must obey Poissonian statistics. Thus, the variance of D_n must equal the ensemble mean of D_n :

$$\text{Var}(D_n) = \langle D_n \rangle \quad (14)$$

The problem is, however, that the ensemble means of the counts in the data bins are unknown. For the bins with very large counts D_n the difference between D_n and its ensemble mean is rather small, which makes it possible to substitute D_n for $\langle D_n \rangle$ in those bins. For the bins with not very large counts D_n such a substitution would result in a significant error, and for the bins with zero counts ($D_n=0$) this substitution would result in a division-by-zero error in equation (12). Clearly, substituting D_n for $\langle D_n \rangle$ does not always work. Note, that if a model adequately fits the data, then the fit M_n is closer to $\langle D_n \rangle$ than the data D_n , which contain random noise. This suggests using the fit M_n for $\text{Var}(D_n)$, however, this approach also has two potential drawbacks: (1) if the fit is inadequate, then the variance estimates $\text{Var}(D_n)$ are incorrect, but these variance estimates are used to calculate the χ^2 that is then used to determine whether the fit is adequate or not, and (2) since the variance estimates $\text{Var}(D_n)$ are dependent on the model, a comparison between two different models based on the χ^2 values would not be fair, because it would be based on different variance estimates. This shows that in order to calculate $\text{Var}(D_n)$, the data D_n must be first fit by a model that (a) adequately fits any lifetime / dwell time distribution and (b) it is not the same model that is being tested whether it fits the data or not. Fortunately, there is such a model that adequately fits any lifetime / dwell time distribution; this model is called the exponential series. Exponential series is a linear combination of a large number of exponential terms with fixed (not adjustable) decay times τ_i that are equally spaced on the logarithmic time scale, with $\tau_{i+1}=1.8 \cdot \tau_i$. The shortest decay times τ_1 equals the width of the first bin ($\tau_1=t_1$), whereas

the longest decay time is close to t_n , where n is the number of the last data bin. The exponential series model is linear in all its adjustable model parameters A_i (since parameters τ_i are not adjustable), therefore this represents a linear least squares problem, which always converges in one step, without iterations. The fit M_n obtained using the exponential series model is used for $\text{Var}(D_n)$ in equations (11-12). This concludes the calculation of the variances $\text{Var}(D_n)$. Once the variances have been calculated, the program Nexp_GLSE switches from the exponential series model to a model that is being tested whether it can provide an adequate fit to the data. This ensures that the calculation of the variances is completely independent of the model being tested and a comparison between different models based on the χ^2 values is fair.

Extensive testing of the above method showed that it gives rather accurate variance estimates for those bins where $\langle D_n \rangle > 0.1$, i. e., there is one or more counts per ten bins on average. For those bins where $\langle D_n \rangle < 0.1$ the variance estimates are not always accurate, which may result in inaccurate χ^2 values and therefore in false conclusions regarding the adequacy of the fit. To avoid these errors the program Nexp_GLSE can truncate the data set (reduce the number of the bins) so that in all remaining bins the M_n (obtained with the exponential series model) is greater than 0.1. This often results in N_D being less than the number of bins requested at the start of the data analysis.

6. Estimates of the standard deviations for the fitting parameters and any functions of them.

A least squares estimator searches for the minimum of the SSQ defined in equation (10). From now on it will be assumed that the weights W_n are defined as in equation (11), therefore there is no difference between the SSQ defined in equation (10) and the χ^2 defined in equation (12). Differentiation of equation (10) with respect to a model parameter p_i results in

$$\frac{\partial \chi^2}{\partial p_i} = -2 \sum_{n=1}^{N_D} W_n [D_n - M_n(\vec{p})] \frac{\partial M_n(\vec{p})}{\partial p_i} \quad (15)$$

At the χ^2 minimum $\partial \chi^2 / \partial p_i$ must be equal to zero, therefore the right-hand side of equation (15) must be equal to zero for every adjustable model parameter p_i . This results in a

system of N_p equations with N_p unknowns. Unless the model function is linear in every parameter p_i , this system of equations cannot be solved algebraically, but the solution can be found using a series of successive iterations. Numerical algorithms that perform the successive iterations can be divided in two categories depending on whether they calculate the Hessian matrix or not. The algorithms that do not calculate the Hessian matrix (common example: the steepest descent algorithm) require an extremely large number of iterations to find the χ^2 minimum; these algorithms are inefficient. The Gauss-Newton algorithm and its modifications, such as the Levenberg-Marquardt algorithm, calculate the Hessian matrix or the matrix $\mathbf{H}$ that will be defined below. In the general case, the Hessian matrix is defined as a square matrix of second-order partial derivatives of a function of multiple variables. In our special case $\frac{1}{2}\chi^2$ plays the role of the function and the parameters p_i play the roles of the variables. The first derivatives of χ^2 can be found in equation (15); dividing these by half and differentiating with respect to p_j yields the second derivatives, which are the elements of the Hessian matrix:

$$\frac{\partial^2 \frac{1}{2}\chi^2}{\partial p_i \partial p_j} = \sum_{n=1}^{N_d} W_n \left(\frac{\partial M_n(\vec{p})}{\partial p_i} \right) \left(\frac{\partial M_n(\vec{p})}{\partial p_j} \right) - \sum_{n=1}^{N_d} W_n [D_n - M_n(\vec{p})] \frac{\partial^2 M_n(\vec{p})}{\partial p_j \partial p_i} \quad (16)$$

The elements of the matrix $\mathbf{H}$ are defined as

$$H_{i,j} = \sum_{n=1}^{N_d} W_n \left(\frac{\partial M_n(\vec{p})}{\partial p_i} \right) \left(\frac{\partial M_n(\vec{p})}{\partial p_j} \right) \quad (17)$$

The relations between the elements of the Hessian matrix and those of the matrix $\mathbf{H}$ are

$$\frac{\partial^2 \frac{1}{2}\chi^2}{\partial p_i \partial p_j} = H_{i,j} - \sum_{n=1}^{N_d} W_n [D_n - M_n(\vec{p})] \frac{\partial^2 M_n(\vec{p})}{\partial p_j \partial p_i} \quad (18)$$

Note, that at the χ^2 minimum the differences $[D_n - M_n(\vec{p})]$ are quite small and randomly change sign, therefore the second term on the right-hand side of equation (18) is expected to vanish at the χ^2 minimum. Compare this with the right-hand side of equation (15), which also contains the differences $[D_n - M_n(\vec{p})]$; at the χ^2 minimum the right-hand side of equation (15) equals zero precisely. Therefore in the vicinity of the χ^2 minimum the Hessian matrix is almost exactly equal to the matrix $\mathbf{H}$. If the model function $M_n(\vec{p})$ is linear in all the fitting parameters p_i , then the Hessian matrix and the matrix $\mathbf{H}$ become completely identical, and

not only in the vicinity of the χ^2 minimum, but everywhere in the parameter space; in the theory of linear least squares the matrix $\mathbf{H}$ is known as "the matrix of the normal equations". In the case of a nonlinear model function, the matrix $\mathbf{H}$ is sometimes called the Hessian matrix, however, according to equation (18), this is only approximately correct and only in the vicinity of the χ^2 minimum. Both the Gauss-Newton and Levenberg-Marquardt algorithm calculate the matrix $\mathbf{H}$ at every iteration, since the elements of this matrix depend on the fitting parameters. During the last iteration the matrix $\mathbf{H}$ is calculated for those values of the fitting parameters that correspond to the χ^2 minimum. If the matrix $\mathbf{H}$ is singular, then the χ^2 minimization algorithm usually fails to converge, however, if the matrix $\mathbf{H}$ is regular, then it can be inverted, and the inverse of the matrix $\mathbf{H}$ will be denoted $\mathbf{C}$. The matrix $\mathbf{C}$ represents the variance-covariance matrix² for the estimates of the fitting parameters p_i . If the ensemble average $\langle p_i \rangle$ is known, then $\delta p_i = p_i - \langle p_i \rangle$ represents the random error in the estimate of the parameters p_i , and the elements of the variance-covariance matrix represent the ensemble averages of the products of the errors:

$$C_{i,j} = \langle \delta p_i \cdot \delta p_j \rangle \quad (19)$$

The diagonal elements ($i=j$) of the variance-covariance matrix represent the variances for the corresponding fitting parameters, therefore the standard deviations can be calculated as

$$\sigma p_i = \sqrt{C_{i,i}} \quad (20)$$

Equation (20) is used to calculate the standard deviations for all *absolute* A_i and τ_i , which play the roles of the fitting parameters in the program Nexp_GLSE. The decay rates k_i , the *fractional* A_i , and both *absolute* α_i and *fractional* α_i are not the fitting parameters, but rather some functions of the fitting parameters, see equations (4-7). For any function φ of the fitting parameters the standard deviation can be calculated as

$$\sigma \varphi = \sqrt{\sum_{i=1}^{N_p} \sum_{j=1}^{N_p} C_{i,j} \left(\frac{\partial \varphi}{\partial p_i} \right) \left(\frac{\partial \varphi}{\partial p_j} \right)} \quad (21)$$

2 Walter Clark Hamilton proved this for the case of a linear model function. In the case of a nonlinear model function this is only approximately correct, therefore the equal sign in equations (19), (20), (21) should be replaced with the approximately equal sign. In the special case of the model function considered here, if the standard deviations for all τ_i are small as compared to the values of the corresponding parameters, then all the standard deviations are accurate, however, if the standard deviation for at least one τ_i is close to or greater than the value of that parameter, then it is likely that multiple standard deviations may be inaccurate.

The program `Nexp_GLSE` calculates the standard deviations using equations (20) and (21) and it also saves the matrix **C** as a file "invhessm.bin"; this is a binary file containing $N_p \times N_p$ REAL(8) numbers. The binary format is used to preserve the full accuracy and also to decrease the size of the file considering that N_p can be a very large number (up to 8192 in the current version of the program).

7. Running the program `Nexp_GLSE` with a single data set.

The program `Nexp_GLSE` can be invoked by typing `Nexp_GLSE` on the command line and pressing the Enter key; in this case the program will write multiple questions on the screen to which the user will have to answer using the keyboard. An example of such dialog is included below; the text typed by the user is highlighted in bold. Alternatively, all the answers can be included in a text file and the program can be invoked by typing `Nexp_GLSE < answer.txt`.

```
dima@xenon:~/for/GitHub/Nexp_GLSE/test_single$ Nexp_GLSE

Number of data sets to be analyzed? 1
1
Number of exponentials? 3
3
Total number of model parameters? 6
6
Number of bins? 3072
3072
Create equally spaced bins or read bin times from File (C/F)? C
C
Bin width? 0.01
1.0000000000000000E-002

Lifetimes / Dwell times file name? simulated_3exp_314.txt
simulated_3exp_314.txt
Is this a List (1-col) or Binned data (2-col)? L
L
10000 lifetimes / dwell times read from file.
8 lifetimes / dwell times are out of range; it is recommended to
increase the number of bins.
Variances were reliably estimated for the first 2050 bins;
the data in remaining bins will be ignored.
Exponential 01: parameter numbers for A and  $\tau$ ? 1, 2
1 2
Exponential 02: parameter numbers for A and  $\tau$ ? 3, 4
3 4
```

Exponential 03: parameter numbers for A and τ ? **5, 6**

5 6

Enter initial guesses for parameters followed by space and T (tunable) or F (fixed).

```

PAR(0001) ? 0.0 T
0.000000E+00 T
PAR(0002) ? 0.01 T
1.000000E-01 T
PAR(0003) ? 0.0 T
0.000000E+00 T
PAR(0004) ? 1.0 T
1.000000E+00 T
PAR(0005) ? 0.0 T
0.000000E+00 T
PAR(0006) ? 10.0 T
1.000000E+01 T

```

Iteration	Chi square	Lambda	Alpha
0000	5.853703613	0.000000	1.000E-08
0001	1.164851501	1.000000	1.000E-08
0002	1.848357531	1.000000	2.500E-09
0003	1.848357464	1.000000	1.000E-08
0004	1.848357197	1.000000	4.000E-08
0005	1.848356131	1.000000	1.600E-07
0006	1.848351863	1.000000	6.400E-07
0007	1.848334794	1.000000	2.560E-06
0008	1.848266522	1.000000	1.024E-05
0009	1.847993497	1.000000	4.096E-05
0010	1.846902421	1.000000	1.638E-04
0011	1.842554466	1.000000	6.554E-04
0012	1.825421877	1.000000	2.621E-03
0013	1.760899246	1.000000	1.049E-02
0014	1.557964127	1.000000	4.194E-02
0015	1.206841712	1.000000	1.678E-01
0016	1.068299192	1.000000	6.711E-01
0017	1.029369731	1.000000	6.711E-01
0018	0.992750302	1.000000	1.678E-01
0019	0.979511433	1.000000	4.194E-02
0020	0.978357199	1.000000	1.049E-02
0021	0.978339663	1.000000	2.621E-03
0022	0.978339656	1.000000	6.554E-04
0023	0.978339656	1.000000	1.638E-04
0024	0.978339656	1.000000	4.096E-05
0025	0.978339656	1.000000	1.024E-05
0026	0.978339656	1.000000	2.560E-06
0027	0.978339656	1.000000	6.400E-07
0028	0.978339656	1.000000	1.600E-07
0029	0.978339656	1.000000	4.000E-08

The χ^2 minimization took 00h 00m 00s 022ms

Reduced χ^2 = 0.978
Number of bins with nonzero weights NB = 2050

Number of free model parameters NP = 6
 Number of degrees of freedom NB-NP = 2044
 The probability of the above χ^2 or greater is 75.398%
 With the probability of 95.0% reduced χ^2 must not exceed 1.052
 With the probability of 99.5% reduced χ^2 must not exceed 1.082
 Absolute χ^2 = 1999.73
 Akaike I.C. = -38.27

Best-fit parameter values with standard deviations:

PAR(0001) = 3.315902E+03 $\pm$ 1.959785E+02
 PAR(0002) = 1.940796E-01 $\pm$ 1.071404E-02
 PAR(0003) = 3.531054E+03 $\pm$ 1.835223E+02
 PAR(0004) = 1.040136E+00 $\pm$ 8.904218E-02
 PAR(0005) = 3.156924E+03 $\pm$ 1.737374E+02
 PAR(0006) = 5.218667E+00 $\pm$ 2.401838E-01

 absolute A(01) = 3.315902E+03 $\pm$ 1.959785E+02
 fraction A(01) = 3.314616E-01 $\pm$ 1.921862E-02
 τ (01) = 1.940796E-01 $\pm$ 1.071404E-02
 k(01) = 5.152524E+00 $\pm$ 2.844417E-01
 absolute α (01) = 1.708526E+04 $\pm$ 5.757357E+02
 fraction α (01) = 8.103045E-01 $\pm$ 1.691575E-02
 absolute A(02) = 3.531054E+03 $\pm$ 1.835223E+02
 fraction A(02) = 3.529685E-01 $\pm$ 1.789266E-02
 τ (02) = 1.040136E+00 $\pm$ 8.904218E-02
 k(02) = 9.614125E-01 $\pm$ 8.230292E-02
 absolute α (02) = 3.394799E+03 $\pm$ 3.514023E+02
 fraction α (02) = 1.610055E-01 $\pm$ 1.561416E-02
 absolute A(03) = 3.156924E+03 $\pm$ 1.737374E+02
 fraction A(03) = 3.155699E-01 $\pm$ 1.726819E-02
 τ (03) = 5.218667E+00 $\pm$ 2.401838E-01
 k(03) = 1.916198E-01 $\pm$ 8.819103E-03
 absolute α (03) = 6.049291E+02 $\pm$ 5.824313E+01
 fraction α (03) = 2.869003E-02 $\pm$ 2.773122E-03

The comment "8 lifetimes / dwell times are out of range; it is recommended to increase the number of bins" can be ignored in some cases but not always. In the above example it followed by "Variances were reliably estimated for the first 2050 bins; the data in remaining bins will be ignored"; the data were originally binned in 3072 bins; 3072/2050=1.498, which shows that increasing the number of bins from 3072 to, for example, 4096 would not make much of a difference. However, if that ratio was less than 1.125 or if the line "the data in remaining bins will be ignored" did not appear at all, then it is strongly recommended to stop the program by pressing Control+C and start over using a greater number of bins (question number 4). The decision how many bins have reliable variance estimates is explained on p. 7 of this document.

In some cases it may be beneficiary to use bins of unequal widths. In order to use this option one must create a file containing all the times t_n (the bin end times) that are defined on p. 3 of this document. It should be an ASCII text file with one number per line; the number of lines must be equal to the number of bins. Answer **F** to the question "Create equally spaced bins or read bin times from File (C/F)?"; in the next question the program will ask for the name of the file containing the times t_n .

The question "Is this a List (1-col) or Binned data (2-col)?" makes it possible to use two types of input data files. If the input file contains unsorted and unbinned data, one leifetime / dwell time per line, then answer either **L** or **1** to this question. However, if the data have been binned already and each line of the data file contains two numbers (separated by a comma and/or any number of spaces but not tabs), the first number being a real number corresponding to the center of the bin, and the second number being an integer equal to the number of counts in this bin, then answer either **B** or **2** to this question. More on the format of the input files in Chapter 9.

When a single data set is analyzed, the choice which parameter number is assigned to each A_i and τ_i is completely free, however, all the parameter numbers must be different. When multiple data sets are analyzed simultaneously, the parameters that apply to multiple data sets will have the same number in all data sets to which they apply.

The choice of the initial guesses for the parameter values and whether they are fixed (F) or tunable (T) is important because it determines whether the χ^2 minimization algorithm will converge or not. In general, depending on whether the model function is a linear function or a nonlinear function of a parameter, that parameter is referred to as a linear or nonlinear parameter. In the case of the model function from equation (2) all A_i are linear parameters and all τ_i are nonlinear parameters. The version of Levenberg-Marquardt algorithm that is used in the program Nexpt_GMLE has one useful feature that other versions may not support: if the user enters zero values as the initial guesses for all linear parameters (for all A_i in the case of the model function considered here), then during iteration 1 the program will find the best possible initial guesses for the linear parameters; then, starting from iteration 2, it will optimize all the parameters, linear and nonlinear. This is a recommended way to choose the initial guesses for the linear parameters if they are all tunable, since this increases the chances that the algorithm will converge; if non-zero guesses are used for the linear parameters and some of them have the wrong sign (positive

instead of negative or negative instead of positive), then the algorithm will either converge to the wrong χ^2 minimum or not converge at all. However, if the value(s) of one or more A_i are known a priori, then that/those value(s) must be used as the initial guess(es) and fixed; in this case non-zero initial guesses must be provided for the tunable A_i as well.

The initial guesses for τ_i must be all positive and very different numbers (close initial guesses for τ_i almost guarantee that the algorithm will not converge). Also the initial guesses for τ_i that are significantly shorter than the width of the first bin t_1 or significantly longer than time at which the last bin ends are recipes for disaster unless the extreme τ_i values are fixed; for fixed parameters τ_i the extreme values are acceptable. If $N_{\text{exp}}=1$, then the algorithm will likely converge regardless of the choice of the initial guess for τ_1 , as long as it is a positive number and not much shorter than t_1 . If $N_{\text{exp}}=2$, then one can take the initial guess for τ_1 close to the width of the first bin t_1 and the initial guess for τ_2 close to the time at which the last bin ends; in most cases this ensures that the algorithm converges without problems. If $N_{\text{exp}}>2$, then the following approach is recommended. First fit the same data with one fewer exponential term, record the τ_i values that resulted in the χ^2 minimum and sort them in the ascending order if they are not in the ascending order already. For the initial guesses with the full number of exponential terms N_{exp} use the following: for the shortest and the longest τ_i use the values that were recommended above for the case $N_{\text{exp}}=2$, and for all intermediate values of τ_i use the geometric mean between the values of τ_{i-1} and τ_i obtained with one fewer exponential term. For example, for $N_{\text{exp}}=3$, use t_1 as the initial guess for τ_1 , $\sqrt{\tau_1 \tau_2}$ as the initial guess for τ_2 and the time at which the last bin ends for τ_3 . This does not guarantee that the algorithm will converge, but the chances are high.

8. Running the program Nexpe_GLSE with multiple data sets (global analysis).

The program Nexpe_GLSE can be used to analyze multiple data sets with some of the fitting parameters being shared between the sets (global analysis). This chapter presents an example of the dialog that takes place in the case of the global analysis, however, all the information in Chapter 7 is also relevant to the global analysis and will not be repeated. Therefore, please read Chapter 7 first. As it was stated in Chapter 7, the program can be invoked

by typing `Nexp_GLSE < answer.txt` (the naming of the text file with the answers is up to the user), which is strongly recommended in the case of the global analysis, since the number of questions is large and if the user makes a typo, then there is no way to go back and the process will have to be started over. If there is a typo in the text file, then the user can correct only the wrong line using an ASCII text editor such as `gedit` in Linux or `TextEdit` in OS X, and then start the program again. Below is a trivial example of global analysis, where three data sets are fit with three-exponential model functions; all A_i are local fitting parameters (not linked between the data sets), whereas some of the τ_i are global fitting parameters and are linked in the following way: τ_1 is linked between data sets 1 and 2, τ_2 is linked between data sets 2 and 3, and τ_3 is linked between data sets 1 and 3. All three data sets have been simulated, each data set contains 10000 dwell times, all A_i for all exponential terms and all data sets were equal to 3333.333, while the values of the decay times were 0.1, 0.5, 3.0 for data set 1; 0.1, 1.0, 5.0 for data set 2; and 0.2, 1.0, 3.0 for data set 3. The dialog associated with the global analysis using the keyboard is shown below; the text typed by the user is highlighted in bold.

```
dima@xenon:~/for/GitHub/Nexp_GLSE/test_global$ Nexp_GLSE

Number of data sets to be analyzed? 3
3
Number of exponentials for one data set (same N for all sets)? 3
3
Total number of model parameters (global+local)? 15
15
Are bins defined identically for all data sets (Y/N)? Y
Y
Number of bins per one data set? 3072
3072
Create equally spaced bins or read bin times from File (C/F)? C
C
Bin width? 0.01
1.0000000000000000E-002

**** Data set 0001 ****
Lifetimes / Dwell times file name? simulated_01.txt
simulated_01.txt
Is this a List (1-col) or Binned data (2-col)? L
L
10000 lifetimes / dwell times read from file.
Variances were reliably estimated for the first 1434 bins;
the data in remaining bins will be ignored.

All parameters are numbered sequentially from 0001 to 0015
Global parameters are those that correspond to multiple data sets.
```

For faster operation global parameters should have smaller numbers.

Exponential 01: parameter numbers for A and τ ? **7, 1**

7 1

Exponential 02: parameter numbers for A and τ ? **8, 4**

8 4

Exponential 03: parameter numbers for A and τ ? **9, 3**

9 3

**** Data set 0002 ****

Lifetimes / Dwell times file name? **simulated_02.txt**

simulated_02.txt

Is this a List (1-col) or Binned data (2-col)? **L**

L

10000 lifetimes / dwell times read from file.

14 lifetimes / dwell times are out of range; it is recommended to increase the number of bins.

Variances were reliably estimated for the first 1994 bins;
the data in remaining bins will be ignored.

Exponential 01: parameter numbers for A and τ ? **10, 1**

10 1

Exponential 02: parameter numbers for A and τ ? **11, 2**

11 2

Exponential 03: parameter numbers for A and τ ? **12, 5**

12 5

**** Data set 0003 ****

Lifetimes / Dwell times file name? **simulated_03.txt**

simulated_03.txt

Is this a List (1-col) or Binned data (2-col)? **L**

L

10000 lifetimes / dwell times read from file.

Variances were reliably estimated for the first 1442 bins;
the data in remaining bins will be ignored.

Exponential 01: parameter numbers for A and τ ? **13, 6**

13 6

Exponential 02: parameter numbers for A and τ ? **14, 2**

14 2

Exponential 03: parameter numbers for A and τ ? **15, 3**

15 3

Enter initial guesses for parameters followed by space and T (tunable) or F (fixed).

PAR(0001) ? **0.01 T**

1.000000E-02 T

PAR(0002) ? **1.00 T**

1.000000E+00 T

PAR(0003) ? **100. T**

1.000000E+02 T

PAR(0004) ? **1.00 T**

1.000000E+00 T

PAR(0005) ? **100. T**

1.000000E+02 T

PAR(0006) ? **0.01 T**

1.000000E-02 T

```

    PAR(0007) ? 0.0 T
0.000000E+00 T
    PAR(0008) ? 0.0 T
0.000000E+00 T
    PAR(0009) ? 0.0 T
0.000000E+00 T
    PAR(0010) ? 0.0 T
0.000000E+00 T
    PAR(0011) ? 0.0 T
0.000000E+00 T
    PAR(0012) ? 0.0 T
0.000000E+00 T
    PAR(0013) ? 0.0 T
0.000000E+00 T
    PAR(0014) ? 0.0 T
0.000000E+00 T
    PAR(0015) ? 0.0 T
0.000000E+00 T

```

Iteration	Chi square	Lambda	Alpha
0000	7.131706117	0.000000	1.000E-08
0001	2.203108108	1.000000	1.000E-08
0002	1.866974516	1.000000	4.194E-02
0003	1.498207640	1.000000	1.678E-01
0004	1.581116586	1.000000	6.711E-01
0005	1.414565022	0.500000	1.000E+00
0006	1.311111444	1.000000	1.000E+00
0007	1.176752533	1.000000	2.500E-01
0008	1.086066063	1.000000	6.250E-02
0009	1.055017507	1.000000	1.000E+00
0010	1.039328803	1.000000	1.000E+00
0011	1.023721888	1.000000	2.500E-01
0012	1.004986824	1.000000	6.250E-02
0013	0.982533643	1.000000	1.562E-02
0014	0.976334617	1.000000	3.906E-03
0015	0.976227057	1.000000	9.766E-04
0016	0.976226934	1.000000	2.441E-04
0017	0.976226933	1.000000	6.104E-05
0018	0.976226933	1.000000	1.526E-05
0019	0.976226933	1.000000	3.815E-06
0020	0.976226933	1.000000	9.537E-07
0021	0.976226933	1.000000	2.384E-07
0022	0.976226933	1.000000	5.960E-08
0023	0.976226933	1.000000	1.490E-08

The χ^2 minimization took 00h 00m 00s 063ms

Reduced χ^2 = 0.976

Number of bins with nonzero weights NB = 4870

Number of free model parameters NP = 15

Number of degrees of freedom NB-NP = 4855

The probability of the above χ^2 or greater is 87.979%

With the probability of 95.0% reduced χ^2 must not exceed 1.034

With the probability of 99.5% reduced χ^2 must not exceed 1.053

Absolute χ^2 = 4739.58
Akaike I.C. = -100.42

Best-fit parameter values with standard deviations:

PAR(0001) = $9.473458\text{E-}02 \pm 2.980208\text{E-}03$
PAR(0002) = $9.384141\text{E-}01 \pm 4.897719\text{E-}02$
PAR(0003) = $3.029535\text{E+}00 \pm 7.894431\text{E-}02$
PAR(0004) = $4.639588\text{E-}01 \pm 2.818757\text{E-}02$
PAR(0005) = $4.889874\text{E+}00 \pm 1.812640\text{E-}01$
PAR(0006) = $1.866115\text{E-}01 \pm 9.781547\text{E-}03$
PAR(0007) = $3.054010\text{E+}03 \pm 1.447767\text{E+}02$
PAR(0008) = $3.513933\text{E+}03 \pm 1.399220\text{E+}02$
PAR(0009) = $3.434976\text{E+}03 \pm 1.130840\text{E+}02$
PAR(0010) = $3.206458\text{E+}03 \pm 9.015027\text{E+}01$
PAR(0011) = $3.367508\text{E+}03 \pm 1.421612\text{E+}02$
PAR(0012) = $3.408863\text{E+}03 \pm 1.391900\text{E+}02$
PAR(0013) = $3.133661\text{E+}03 \pm 1.639201\text{E+}02$
PAR(0014) = $3.550558\text{E+}03 \pm 2.232393\text{E+}02$
PAR(0015) = $3.310422\text{E+}03 \pm 1.698383\text{E+}02$

Best-fit parameter values for data set 0001

absolute A(01) = $3.054010\text{E+}03 \pm 1.447767\text{E+}02$
fraction A(01) = $3.053119\text{E-}01 \pm 1.412781\text{E-}02$
 $\tau(01)$ = $9.473458\text{E-}02 \pm 2.980208\text{E-}03$
 $k(01)$ = $1.055581\text{E+}01 \pm 3.320699\text{E-}01$
absolute $\alpha(01)$ = $3.223754\text{E+}04 \pm 1.177694\text{E+}03$
fraction $\alpha(01)$ = $7.873343\text{E-}01 \pm 1.647987\text{E-}02$
absolute A(02) = $3.513933\text{E+}03 \pm 1.399220\text{E+}02$
fraction A(02) = $3.512908\text{E-}01 \pm 1.351567\text{E-}02$
 $\tau(02)$ = $4.639588\text{E-}01 \pm 2.818757\text{E-}02$
 $k(02)$ = $2.155364\text{E+}00 \pm 1.309480\text{E-}01$
absolute $\alpha(02)$ = $7.573804\text{E+}03 \pm 6.028831\text{E+}02$
fraction $\alpha(02)$ = $1.849743\text{E-}01 \pm 1.567653\text{E-}02$
absolute A(03) = $3.434976\text{E+}03 \pm 1.130840\text{E+}02$
fraction A(03) = $3.433974\text{E-}01 \pm 1.082827\text{E-}02$
 $\tau(03)$ = $3.029535\text{E+}00 \pm 7.894431\text{E-}02$
 $k(03)$ = $3.300837\text{E-}01 \pm 8.601398\text{E-}03$
absolute $\alpha(03)$ = $1.133830\text{E+}03 \pm 6.010386\text{E+}01$
fraction $\alpha(03)$ = $2.769141\text{E-}02 \pm 1.608403\text{E-}03$

Best-fit parameter values for data set 0002

absolute A(01) = $3.206458\text{E+}03 \pm 9.015027\text{E+}01$
fraction A(01) = $3.211973\text{E-}01 \pm 8.414902\text{E-}03$
 $\tau(01)$ = $9.473458\text{E-}02 \pm 2.980208\text{E-}03$
 $k(01)$ = $1.055581\text{E+}01 \pm 3.320699\text{E-}01$
absolute $\alpha(01)$ = $3.384676\text{E+}04 \pm 9.744432\text{E+}02$
fraction $\alpha(01)$ = $8.876117\text{E-}01 \pm 5.533724\text{E-}03$
absolute A(02) = $3.367508\text{E+}03 \pm 1.421612\text{E+}02$
fraction A(02) = $3.373300\text{E-}01 \pm 1.371188\text{E-}02$
 $\tau(02)$ = $9.384141\text{E-}01 \pm 4.897719\text{E-}02$
 $k(02)$ = $1.065628\text{E+}00 \pm 5.561664\text{E-}02$
absolute $\alpha(02)$ = $3.588510\text{E+}03 \pm 1.884681\text{E+}02$
fraction $\alpha(02)$ = $9.410660\text{E-}02 \pm 5.130739\text{E-}03$
absolute A(03) = $3.408863\text{E+}03 \pm 1.391900\text{E+}02$

```

fraction A(03) = 3.414726E-01 ± 1.364875E-02
      τ(03) = 4.889874E+00 ± 1.812640E-01
      k(03) = 2.045042E-01 ± 7.580822E-03
absolute α(03) = 6.971270E+02 ± 5.047120E+01
fraction α(03) = 1.828175E-02 ± 1.388084E-03

```

```

Best-fit parameter values for data set 0003
absolute A(01) = 3.133661E+03 ± 1.639201E+02
fraction A(01) = 3.135341E-01 ± 1.610094E-02
      τ(01) = 1.866115E-01 ± 9.781547E-03
      k(01) = 5.358725E+00 ± 2.808863E-01
absolute α(01) = 1.679243E+04 ± 5.860256E+02
fraction α(01) = 7.749618E-01 ± 1.361575E-02
absolute A(02) = 3.550558E+03 ± 2.232393E+02
fraction A(02) = 3.552462E-01 ± 2.201027E-02
      τ(02) = 9.384141E-01 ± 4.897719E-02
      k(02) = 1.065628E+00 ± 5.561664E-02
absolute α(02) = 3.783573E+03 ± 3.077899E+02
fraction α(02) = 1.746099E-01 ± 1.354575E-02
absolute A(03) = 3.310422E+03 ± 1.698383E+02
fraction A(03) = 3.312197E-01 ± 1.671309E-02
      τ(03) = 3.029535E+00 ± 7.894431E-02
      k(03) = 3.300837E-01 ± 8.601398E-03
absolute α(03) = 1.092716E+03 ± 7.836620E+01
fraction α(03) = 5.042829E-02 ± 3.905309E-03

```

The parts of the dialog that are specific to global analysis (where the number of sets is greater than one) are explained below; for the explanation of the parts that are common with single-data-set analysis see Chapter 7. Whether the bins are of equal width or not, it is possible that all data sets involved in the analysis have the same number of bins and the same bin end times t_n . In this case the user can define the bins only once by answering "Y" to the question "Are bins defined identically for all data sets (Y/N)?". If the answer to this question is "N", then the program will ask to define bins for every data set individually.

"All parameters are numbered sequentially from 0001 to 0015. Global parameters are those that correspond to multiple data sets. For faster operation global parameters should have smaller numbers." - this is a brief reminder how the model parameters should be numbered. A global parameter is involved in multiple data sets, and it must have the same number for those data sets. A local parameter is involved in only one data set and it cannot have the same number as a local parameter from another data set or a global parameter. The suggestion that global parameters should have smaller numbers makes a lot of sense when the total number of model parameters is very large (hundreds or thousands), because the inversion of the matrix $\mathbf{H}$ (defined

in Chapter 6) that is carried out at every iteration takes less time when local parameters follow after the global parameters, however, for small numbers of model parameters, like 15 in this example, the difference in speed is insignificant. In any case, the order in which the parameters are numbered should have no effect on the accuracy of the results.

In this specific example, the global parameters are τ_1 for data sets 1 and 2, τ_2 for data sets 2 and 3, and τ_3 for data sets 1 and 3; these fitting parameters were assigned the parameter numbers 1, 2, and 3, respectively. The rest of the τ_i are local parameters, they were assigned the numbers 4, 5, and 6. There are 9 local parameters A_i (three per data set), they were assigned numbers 7 through 15. This explains the logic of assigning parameter numbers to the actual fitting parameters.

The initial guesses for the fitting parameters should be chosen using the guidance in Chapter 7, however, in the case of the global analysis the initial guesses may be somewhat less important than in the case of a single-data-set analysis: the initial guesses that would result in no convergence of the χ^2 minimization algorithm with a single data set may result in convergence of that algorithm in the case of the global analysis.

9. Input data files.

There are three kinds of input files that the program Nexp_GLSE will read. The user can name the input files as he/she likes; the program will ask for the name of each input file when it needs to read it. The names of the input files can include the full path (if these files are located in a directory other than the current directory in which the program is invoked). All input files are ASCII text files.

1. "file with bin end times" - the program will ask for that file if the user wants to use bins of unequal width. The file must contain one line per bin, each line must contain one real number, and this number must be equal to t_n , where t_n is the end time of the bin n , as defined in Chapter 3. Bin number 1 begins at $t=0$ and ends at $t=t_1$ (the number in the first line of the file). Bin number n begins at $t=t_{n-1}$ and ends at $t=t_n$. The numbers t_n must be all positive and they must represent a monotonically increasing sequence, i. e., $t_n < t_{n+1}$. A lifetime / dwell time t is placed in such bin n that $t_{n-1} < t \leq t_n$. Example:

```

0.01
0.02
0.04
0.08
0.16
...

```

2. A list of lifetimes / dwell times (1-column data format) - this is the first alternative of the input data format. A separate input file must be created for each data set. The file must contain one line per lifetime / dwell time, one real number per line. Example:

```

24.550235657838247
1.6578485980036026
2.0973158069636080
15.138800907719274
1.7881319697327462
0.16318530363790668
4.0418479856398307E-002
3.7954658449139269
.....

```

3. Binned data (2-column data format) - the other alternative of the input data format. A separate input file must be created for each data set. Each line in this file corresponds to one bin (rather than one dwell time) and must contain two numbers. The first must be a real number equal to any time t that satisfies the relation $t_{n-1} < t \leq t_n$ for the given bin n . Considering possible errors in converting decimal fractions to binary, it is not recommended to use t values close to t_{n-1} or t_n ; the safest choice is $(t_{n-1} + t_n)/2$. The second number must be an integer D_n , which is the count of dwell times in bin n . If the second number is a real number rather than an integer, then the program will still be able to read this format, but it will round the number to the nearest integer. The separator between the first and the second number in an input file can be a comma and/or any number of spaces (but not tabs). The order of lines in this file can be random (it is not required that the numbers in the first column would be sorted in ascending or descending order). The lines corresponding to the bins with zero counts D_n can be skipped. Example:

```

5.000000000000000E-003 198
1.500000000000000E-002 202
2.500000000000000E-002 203
3.500000000000000E-002 200
4.500000000000000E-002 159
5.500000000000000E-002 162
6.500000000000000E-002 174
7.500000000000000E-002 125
8.500000000000000E-002 143
.....

```

10. Output files.

For each data analysis the program Nexp_GLSE generates a number of files with fixed names in the current directory. The file names are results.txt, chisquare.txt, parameters.txt, invhessm.bin, binned_####@.txt, expser_####@.txt, Nexffit_####@.txt, and WghResid_####@.txt, where #### denotes four digits (from 0001 to 9999) that represent the number of the data set and @ denotes a letter B (for Bar graph) or D (for Dots).

The file results.txt is a text file, it uses UTF-8 encoding and intended for reading by the user. It contains a copy of the most important information that was printed on the screen during the data analysis, including the information on the input data, the numbering of the model parameters, the initial guesses for the model parameters and their final values with standard deviations, etc.. In other words, this is a detailed record of what was done.

The files chisquare.txt and parameters.txt are ASCII text files that are convenient for reading by another program. The file chisquare.txt contains:

- the minimized *reduced* χ^2 value,
- the total number of data bins in all data sets N_D ,
- the number of free fitting parameters N_P ,
- the number of degrees of freedom $N_F = N_D - N_P$,
- the probability P such that the *reduced* $\chi^2(1-P, N_F)$ equals the *reduced* χ^2 above,
- the table value of *reduced* $\chi^2(0.95, N_F)$,
- the table value of *reduced* $\chi^2(0.995, N_F)$,
- the minimized χ^2 value (absolute rather than reduced),
- the Akaike Information Criterion $AIC = 2N_P + \chi^2 - N_D$,
- either T if the algorithm converged or F if it did not.

The file parameters.txt contains one line per each model parameter with two numbers per line: the best-fit value and the standard deviation for the parameter.

The binary file invhessm.bin contains $N_P \times N_P$ REAL(8) numbers, the elements of the matrix **C**, see Chapter 6. The binary format preserves the full accuracy and reduces the file size.

The files binned_####@.txt, expser_####@.txt, Nexffit_####@.txt, and WghResid_####@.txt are intended as input files for a plotting program. Each of these files contains two columns of real numbers. Abscissa values in the first column always represent the time t . Ordinate values in the second column represent the following quantities. The files binned_####@.txt contain binned experimental data D_n . The files expser_####@.txt contain the fits by the model of exponential series, which are used to calculate the variances $Var(D_n)$ and

weights W_n for each bin, see Chapter 5. The files Nexffit_####@.txt contain the fits $M_n(\vec{p})$ by the selected model of N_{exp} exponential terms. The files WghResid_####@.txt contain the weighted residuals. Each of the data files is presented in two formats, one with a letter B (for Bar graph) and one with the letter D (for bin-centered Dots) in place of @ in the file name; the difference can be clearly seen in Fig. 1.

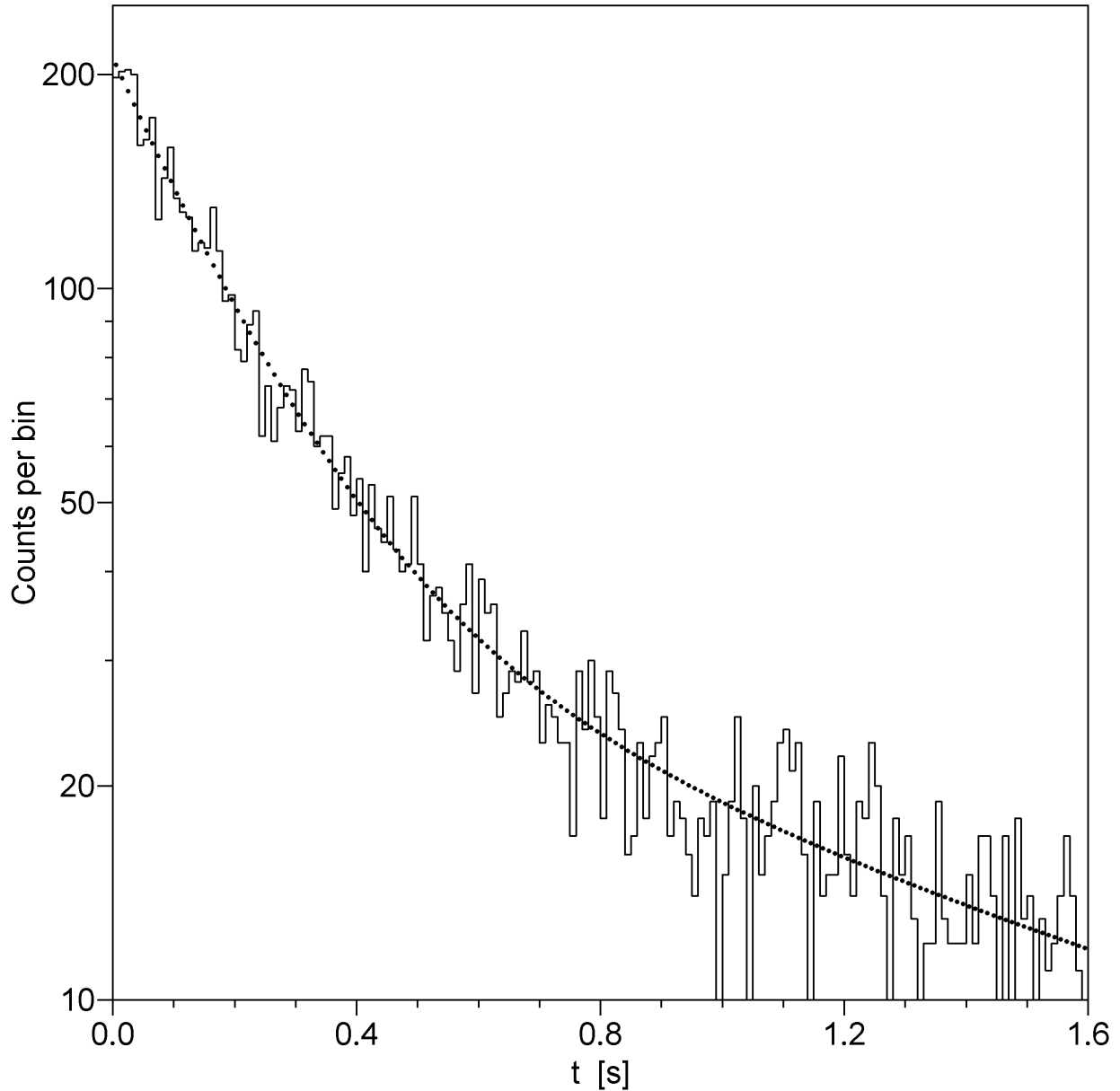


Figure 1. A semi-log plot of the data and fit from the example in Chapter 7, where the solid line (a bar graph) represents the binned data from the file binned_0001B.txt, and the round dots represent the model fit from the file Nexffit_0001D.txt.

The weighted residuals are calculated as

$$\text{Weighted Residual}_n = \sqrt{W_n} \cdot [D_n - M_n(\vec{p})] = \frac{D_n - M_n(\vec{p})}{\sqrt{\text{Var}(D_n)}} \quad (22)$$

and represent a good alternative to the χ^2 for subjectively judging the quality of the fit. If the counts in all bins are very large and the fit is perfect, then the weighted residuals are expected to look something like those in Fig. 2.

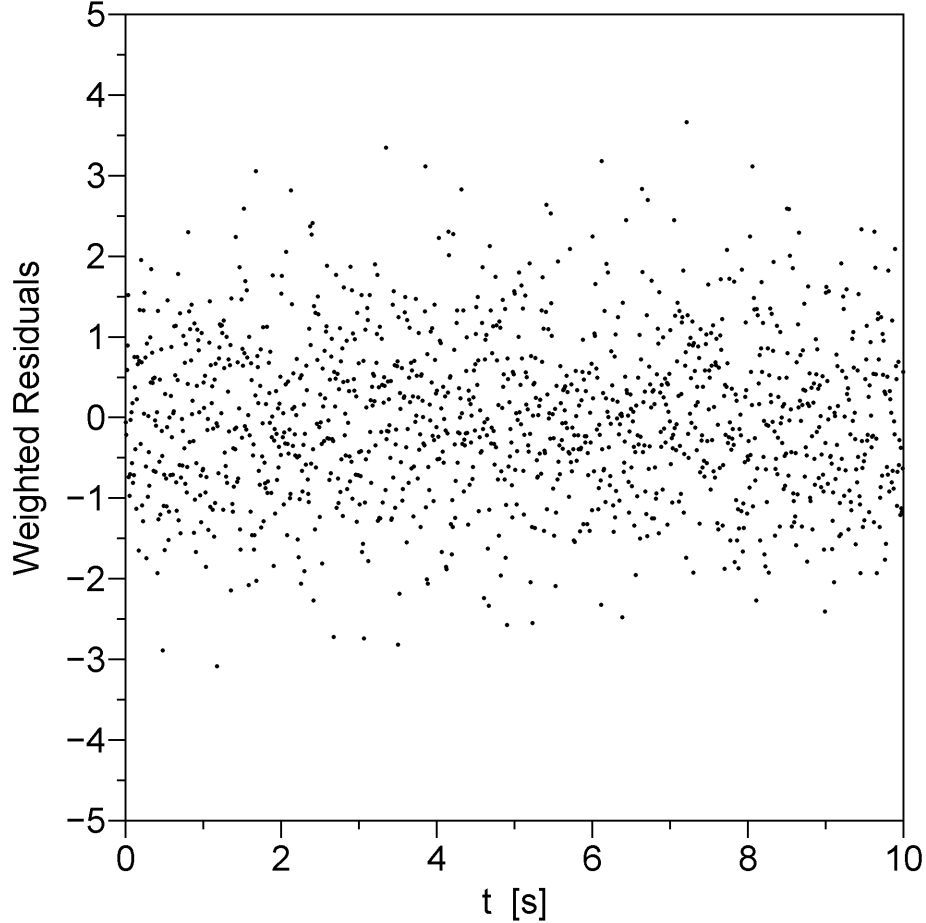


Figure 2. An example of weighted residuals (file WghResid_0001D.txt) for the case of very large counts in all bins and a perfect fit.

Note, that the weighted residuals in Fig. 2 fall on a horizontal strip of approximately constant width from left to right. This is an important feature of weighted residuals; if the residuals were not weighted, then the strip would be wider on the left and narrower on the right. If the number of counts in the bins are low (which often occurs at late times), then differences between a Gaussian distribution (where the random variable can assume any real value) and a Poissonian

distribution (where the random variable can assume only integer values) becomes apparent in weighted residual plots, which is clearly visible in Fig. 3. Note, that at $t > 2$ s the dots in Fig. 3 fall on curved lines, where each line corresponds to a specific number of dwell time counts.

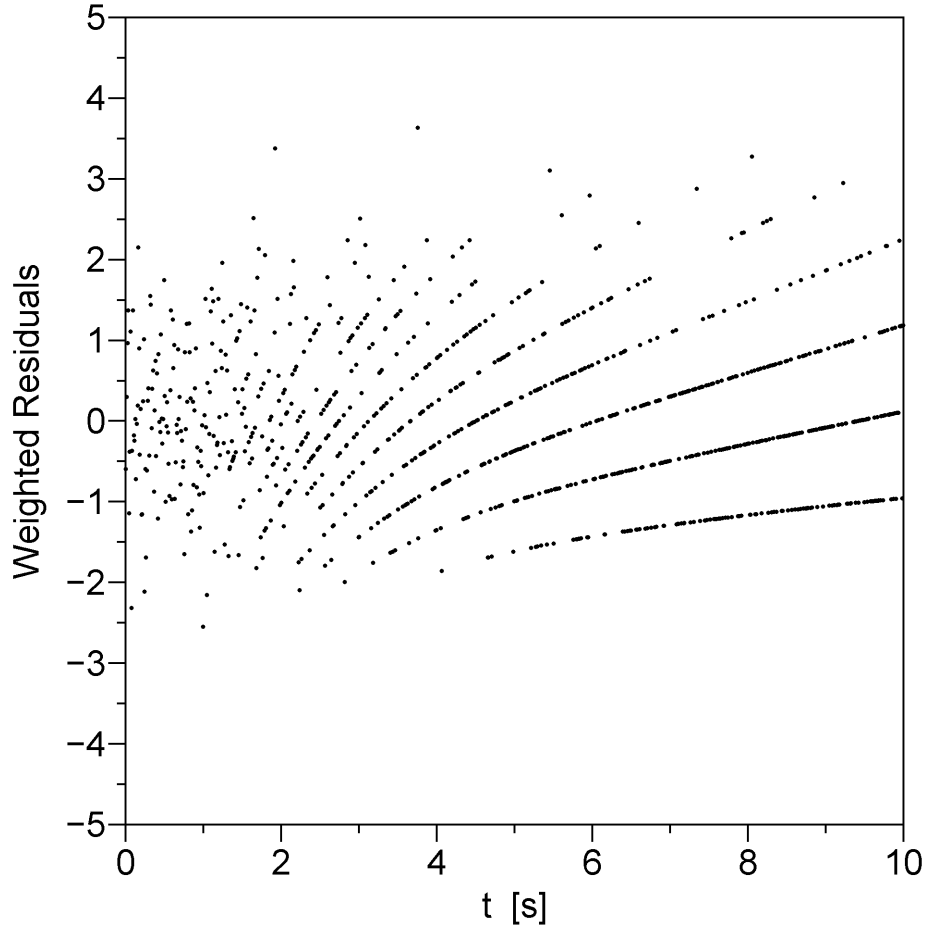


Figure 3. An example of weighted residuals (file WghResid_0001D.txt) for the case of low counts in some bins and a perfect fit.

Note, that in Fig. 3, as well as in Fig. 2, the great majority of the dots fall between the values of -3.0 and $+3.0$ on the vertical axis. This indicates a good fit.

An example of a bad fit is presented in Fig. 4. The the data were simulated as a sum of three exponential terms, with $\tau_1=0.2$ s, $\tau_2=1.0$ s, $\tau_3=5.0$ s, one million total dwell times were simulated and were fit with a model of two exponential terms. The dots in Fig. 4 do not fall between the values of -3.0 and $+3.0$ on the vertical scale, and if one manually draws a "fit" line through the highest density of these dots, then this line will cross $Y=0$ four times; the number of

zero crossings being equal to the number of the adjustable fitting parameters in the model that resulted in an inadequate fit to the data.

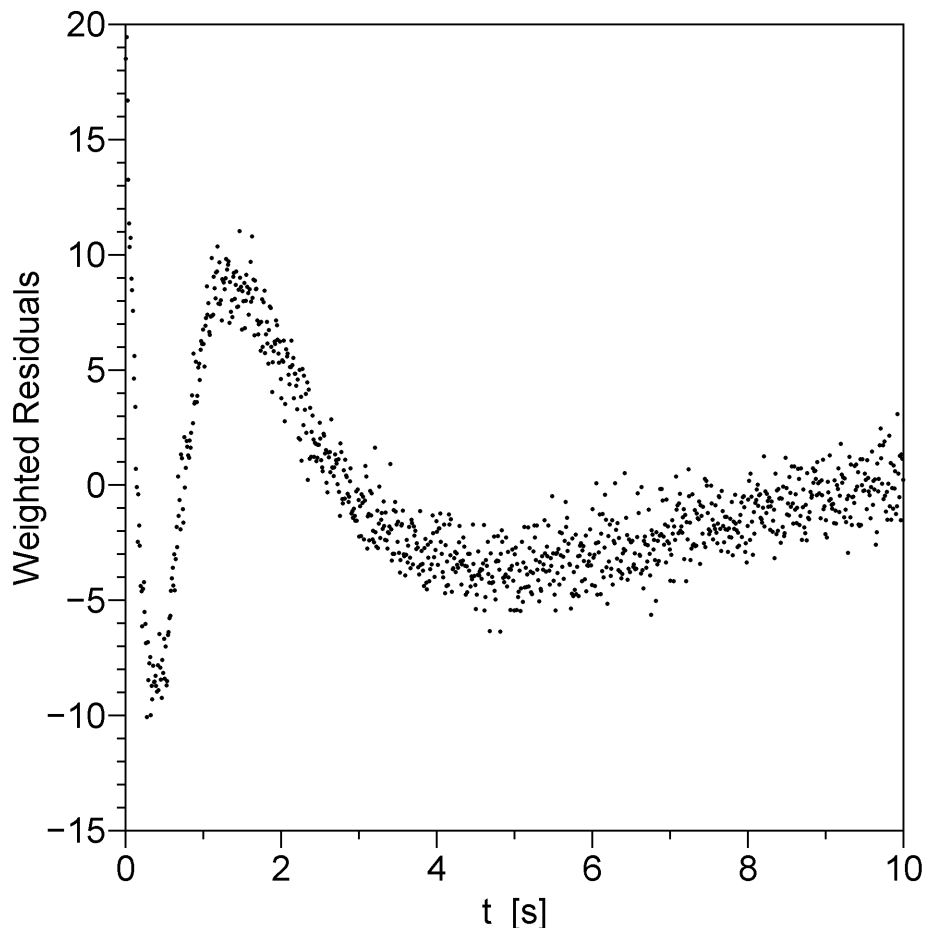


Figure 4. An example of weighted residuals (file WghResid_0001D.txt) for the case of high counts in all bins and an inadequate fit (2-exponential fit to 3-exponential simulated data).

11. Dwell time simulator program for testing purposes.

Dwell time data for the analysis by the program Nexp_GMLE can be simulated using the program Nexp_dwelltime_simulator. The dwell time simulator program can be invoked by typing Nexp_dwelltime_simulator on the command line and pressing the Enter key; in this case the program will write multiple questions on the screen to which the user will have to answer using the keyboard. An example of such dialog is included below; the text typed by the user is

highlighted in bold. Alternatively, all the answers can be included in a text file and the program can be invoked by typing `Nexp_dwelltime_simulator < answer.txt` (the naming of the text file with the answers is up to the user).

```
dima@xenon:~/for/GitHub/Nexp_GLSE/test_simulator$ Nexp_dwelltime_simulator

Number of exponentials? 3
3
A(1),  $\tau(1)$  ? 1.0, 1.0
1.0000000000000000 1.0000000000000000
A(2),  $\tau(2)$  ? 1.0, 5.0
1.0000000000000000 5.0000000000000000
A(3),  $\tau(3)$  ? 1.0, 25.0
1.0000000000000000 25.0000000000000000
Number of dwell times? 10000
10000
Random number generator seed ( 8 integer numbers)? FOOBAR
1582331974 -267422310 -1258900642 679466304 -1461064602 -1502387896
-1942932215 1069698660
Output file name? simulated_dwelltimes.txt
simulated_dwelltimes.txt
STOP OK.
```

The above is an example of a simulation using a triple-exponential survival function ($N_{\text{exp}}=3$), where all three amplitudes A_i are equal to one another, $\tau_1=1.0$, $\tau_2=5.0$, $\tau_3=25.0$, a total of 10000 dwell times are generated and saved in the file "simulated_dwelltimes.txt". The amplitudes A_i can be either absolute or fractional; since the program re-normalizes the input numbers A_i there is no difference whether absolute or fractional amplitudes are used. The question regarding the seed for the random number generator requires one of the following: either eight integer numbers or any text that is not a number; in the latter case the program will suggest eight integer numbers and print them on the screen. Using the same random seed numbers results in the same set of generated dwell times, which may be needed for reproducibility of the results in some tests.

12. Suppression of systematic errors.

The program `Nexp_GLSE` was extensively tested using simulated dwell time data and was found to produce adequate fits (judged by the χ^2 goodness-of-fit test) in all tests without exception. However, when fitting binned experimental dwell times, the fit was not always adequate. In one application a statistically unacceptable deviation was observed in bin number 1.

Bin number 1 contained very short dwell times; for a short dwell time the observed signal often fell below the detection threshold, therefore the number of dwell times in bin number 1 was underestimated, resulting in a systematic error. To minimize the effect of this systematic error on the results, the data for bin number 1 had to be excluded from the analysis. Toward this goal a new feature was introduced in the program Nexp_GLSE: now the program can effectively exclude specific bins from the data analysis by setting their weights W_n to zero instead of $1/\text{Var}(D_n)$; bins with zero weights are not included in the total count of data bins N_D that is used in calculating the *reduced* χ^2 and AIC.

The list of the bins with systematic errors that need to be excluded from the analysis can be provided to the program Nexp_GLSE in one of the following two ways. In the case where a single data set is analyzed, the user must place in the current directory (in which Nexp_GLSE will be run) an ASCII text file excluded_bins.txt. The number of lines in this file must be equal to the number of excluded bins, with one positive integer per line representing the number of the bin to be excluded. For example, if only bin number 1 needs to be excluded, then the file excluded_bins.txt must contain one line with number 1 in it. In the case where multiple data sets are analyzed globally and the same bin numbers need to be excluded from all data sets, it is sufficient to create one file excluded_bins.txt in the current directory, and it will apply to all data sets. However, if different bin numbers need to be excluded from different data sets, then the user must create multiple files excluded_bins_####.txt, one per data set, where #### denotes the data set number (0001, 0002, 0003, etc.). Each file excluded_bins_####.txt applies to one data set only and contains the list of all bin numbers that need to be excluded, one number per line. The files excluded_bins.txt and/or excluded_bins_####.txt must be created in the current directory (in which Nexp_GLSE will be run) before the program Nexp_GLSE is invoked. If both files excluded_bins.txt and excluded_bins_####.txt exist, then the program Nexp_GLSE will use the file excluded_bins_####.txt. For example, if three data sets are analyzed globally, and in the current directory there are files excluded_bins.txt and excluded_bins_0002.txt, then the program will use the bin numbers from the file excluded_bins.txt for data sets 0001 and 0003, but for the data set 0002 it will use the bin numbers from the file excluded_bins_0002.txt. If no bins need to be excluded from the data set number ####, then there should be neither file excluded_bins_####.txt nor file excluded_bins.txt in the current directory.

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